

%% 题一

```
A=[0.78 -0.02 -0.12 -0.14;-0.02 0.86 -0.04
0.06;
    -0.12 -0.04 0.72 -0.08;-0.14 0.06 -0.08
0.74];
b=[0.76;0.08;1.12;0.68];
n=size(b,1);
D=diag(diag(A));L=-(tril(A)-D);U=-(triu(A)-D);
Mj=D\ (L+U);fj=D\b;
Mgs=(D-L)\U;fgs=(D-L)\b;
omega=1.03;Mso=(D-omega*L)\ ((1-
omega)*D+omega*U);
fso=omega*( (D-omega*L)\b);
x=zeros(n,1);
err=1;k=0;
while err>1e-6
    k=k+1;
    xold=x;
    %      x=Mj*x+fj;%% Jacobi迭代
    %      x=Mgs*x+fgs;%% Gauss-Seidel迭代
    x=Mso*x+fso;%% SOR迭代
    err=norm(x-xold,inf);
end
```

%% 题二

```
A=[0.78 -0.02 -0.12 -0.14;-0.02 0.86 -0.04
0.06;
    -0.12 -0.04 0.72 -0.08;-0.14 0.06 -0.08
0.74];
b=[0.85653;0.42076;-0.23948;-0.60632];
n=size(b,1);
D=diag(diag(A));L=-(tril(A)-D);U=-(triu(A)-D);
Mj=D\ (L+U);fj=D\b;
Mgs=(D-L)\U;fgs=(D-L)\b;
inx=[];
for omega=[0.6 0.8 1 1.1 1.15 1.25 1.3 1.5 1.8]
```

```

Mso=(D-omega*L)\((1-omega)*D+omega*U);
fso=omega*((D-omega*L)\b);
x=zeros(n,1);
err=1;k=0;
while err>1e-10
    k=k+1;
    xold=x;
    %    x=Mj*x+fj;%% Jacobipü'ú
    %    x=Mgs*x+fgs;%% Gauss-Seidelpü'ú
    x=Mso*x+fso;%% Gauss-Seidelpü'ú
    err=norm(x-xold,inf);
end
inx=[inx k];
end

```

%% 题三

```

A=[0.78 -0.02 -0.12 -0.14;-0.02 0.86 -0.04
0.06;
-0.12 -0.04 0.72 -0.08;-0.14 0.06 -0.08
0.74];
n=size(A,2);x=zeros(n,1);
b=[0.76;0.08;1.12;0.68];
r=b-A*x;
err=1;k=0;
while err>1e-6
    k=k+1;
    alpha=r.'*r/(r.'*A*r);
    x=x+alpha*r;
    r=b-A*x;
    err=norm(r,inf)
end

```